REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 11

Capstone Project Planning and Mission Definition

From a Blank Page to a Defined, Bounded, Verifiable Mission

Prepared for	Cao Khanh Tan
Module	11, the planning phase of the capstone
Draws on	The whole programme; feeds the capstone build
Deliverable	A mission definition document (project proposal)
Effort	About 1 week at 12 to 15 hours

SESSION 11

Capstone Project Planning and Mission Definition

Before any trajectory is computed or any constellation is sized, a mission has to be defined. This is the phase that separates a vague idea from an engineering project: deciding what the mission is for, writing down what it must achieve in words that can be verified, bounding it with budgets and constraints, and planning how it will be built and checked. This session walks through that definition process and produces the first capstone deliverable, a mission definition document. Do it well and the rest of the capstone becomes execution; do it poorly and you will design the wrong thing precisely.

Learning objectives

- Write a mission statement and a short set of measurable mission objectives.
- Derive and write verifiable requirements, and trace them through a flowdown.
- Capture a concept of operations across the mission phases.
- Define the trade space, the figures of merit, and the technical budgets and margins.
- Identify and rank the main risks.
- Assemble a mission definition document that scopes the capstone.

> **The phase before the physics**
> Real missions spend a large fraction of their effort here, in concept and definition, long before detailed design. A requirement written badly, or a constraint left unstated, is the most expensive kind of mistake because everything downstream inherits it. Slow down and get the definition right.

PART 1

Mission Statement and Objectives

Start with one or two sentences that say what the mission does and for whom, the mission statement. It is the anchor everything else traces back to. From it flow a handful of mission objectives: the concrete, measurable things the mission must accomplish. Objectives are not yet requirements; they are the "what and why" that requirements will make precise.

> **Example mission statement and objectives (constellation track)**
> **Statement:** Provide continuous, low-latency Earth-observation coverage of all populated latitudes for a seven-year operational life.
> **Objective 1:** Image any point below 60 degrees latitude with a mean revisit under 30 minutes.
> **Objective 2:** Keep the whole system within a defined satellite count and launch-mass budget.
> **Objective 3:** Dispose of every satellite safely within five years of end of life.

PART 2

Requirements: Write Them So They Can Be Verified

A requirement is a "shall" statement that is specific, measurable and, above all, verifiable: there must be a clear way to prove it is met. Requirements flow down in a hierarchy, each one tracing to a parent above it and each one verified from the bottom up.



Figure 1. The requirements flowdown. One mission objective expands into several mission requirements, then system, then subsystem requirements. Traceability runs top down (why does this exist?); verification runs bottom up (is it met?).

A good requirement states a single need, a number, and a condition. Compare a weak and a strong version:

Weak (unverifiable)	Strong (verifiable)
The system shall provide good coverage.	The system shall provide a mean revisit time below 30 min for all latitudes under 60 deg.
The satellites shall be reliable.	Each satellite shall maintain its orbit within a 5 km dead-band for 7 years.
Disposal shall be timely.	Each satellite shall re-enter within 5 years of end of mission.

Every requirement also needs a verification method: analysis, simulation, inspection or test. In the capstone, most requirements are verified by analysis and simulation, and the verification plan is simply the table that says how each one will be shown to hold.

PART 3

Concept of Operations

The concept of operations (ConOps) tells the story of the mission across its life: how it is launched, deployed, operated and disposed of. Writing it early surfaces hidden requirements, such as the delta-v for deployment or the propellant reserved for disposal, that a static design would miss.



Figure 2. A typical mission ConOps. Each phase has its own needs: ascent and deployment set the launch and phasing budget, commissioning checks the system, operations is the mission life, and disposal closes it out. The astrodynamics of each phase comes from the earlier modules.

PART 4

Trade Space, Budgets and Margins

Definition also means deciding what is free to change and what must be tracked. The trade space is the set of design variables (altitude, inclination, satellite count, transfer strategy) and the figures of merit used to compare options (delta-v, revisit, cost). Alongside it sit the technical budgets, running totals of a scarce resource that must not be exceeded: a delta-v budget, a mass budget, a power or link budget. Each carries a margin, the headroom between what the design delivers and what the requirement demands:

$$\text{Margin} = \frac{\text{capability} - \text{requirement}}{\text{requirement}} \times 100\%$$

and budgets are rolled up with a contingency so that early, uncertain estimates do not blow the limit later:

$$\text{Total} = \left(\sum_i x_i\right)(1 + \text{contingency})$$

Example delta-v budget (constellation track, 1200 km)
A simple lifetime delta-v budget for one satellite of the example constellation makes the scope concrete:

Deployment and phasing: ~100 m/s. Station-keeping (7 years): ~40 m/s. Disposal (deorbit from 1200 km): ~300 m/s. Subtotal ~440 m/s, times a 20 percent contingency gives a budget of about 530 m/s per satellite. That single number drives the propellant and therefore the mass.

PART 5

Risk Identification

Finally, name what could go wrong. Each risk is scored by how likely it is and how bad the consequence would be, and the product places it on a risk matrix that tells you where to spend attention:

$$R = L \times C \quad \text{(likelihood} \times \text{consequence)}$$

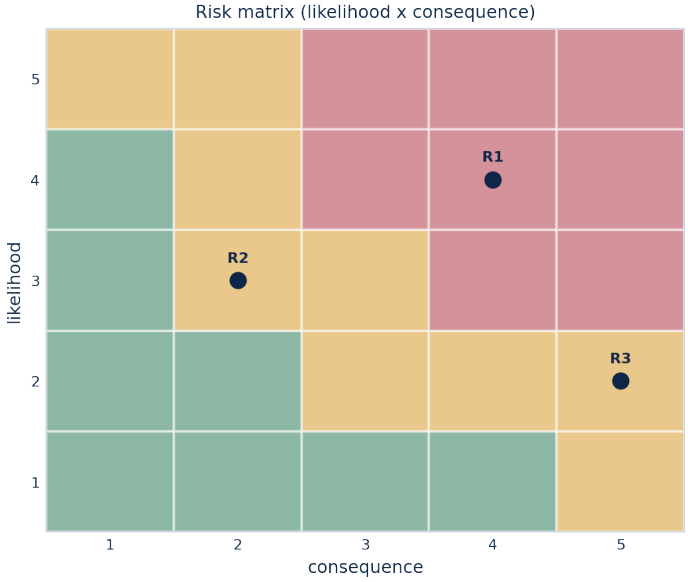


Figure 3. A risk matrix. Likelihood runs up, consequence across; the product sorts risks into low (green), medium (amber) and high (red). Red risks need a mitigation plan before the design proceeds; green ones are simply watched.

Example risks (constellation track)
R1, a launch-vehicle shortfall leaving the constellation incomplete (likely, serious): high, needs a phased-deployment plan. R2, higher-than-modelled drag shortening life (possible, moderate): medium, carry extra station-keeping margin. R3, a single satellite failure degrading coverage (unlikely, serious): accepted with a spare in each plane.

The Mission Definition Document

The output of this phase is a single document that any reviewer could read and understand the mission from. It gathers everything above and becomes the opening chapters of the capstone report. Its structure is the checklist on the reference page.

How this feeds the capstone
Sessions 1 to 10 gave you the methods; this session gives you the mission to point them at. With a defined mission, the capstone build (design, optimise, simulate, verify, defend) is execution against your own requirements. Bring the definition document to the next session as your proposal.

PRACTICE

This Week: Define Your Capstone Mission (12 to 15 hours)

This week has one deliverable: a mission definition document for the capstone track you choose. Work through the steps in order.

Deliverable: Mission Definition Document

- **Choose a track and write the statement (2 to 3 h).** Pick one of the capstone tracks (interplanetary, constellation, orbit maintenance or rendezvous). Write a one or two sentence mission statement and three measurable objectives.
- **Requirements (3 to 4 h).** Write at least six verifiable "shall" requirements, each tracing to an objective, and give each a verification method (analysis, simulation, inspection). Draw the flowdown.
- **ConOps (2 h).** Sketch the concept of operations across the mission phases, noting the astrodynamics need at each phase.
- **Trade space and budgets (3 to 4 h).** List the design variables and figures of merit, and build one technical budget (a delta-v or mass budget) with a contingency and a stated margin.
- **Risk and assembly (2 h).** Identify at least three risks, place them on a risk matrix, and assemble everything into a mission definition document.

What good looks like

Element	A strong submission has
Objectives	measurable, each with a number
Requirements	six or more verifiable shall-statements, each traced and with a verification method
ConOps	all mission phases, with the astrodynamics need named at each
Budget	one technical budget with contingency and a computed margin
Risk	at least three scored risks on a matrix, with a mitigation for any red

Coming up in Session 12: the Capstone Build

With the mission defined, the final session is execution: design and optimise the baseline, simulate and verify it against your own requirements, write the full mission report, and defend it. The methods of Sessions 1 to 10, pointed at the mission you defined this week.

REFERENCE

Mission Definition Document: Template and Tools

Structure your mission definition document in these sections:

- **1. Mission statement and objectives** - what, for whom, and the measurable goals.
- **2. Requirements** - the verifiable shall-statements, with the flowdown and a traceability note.
- **3. Concept of operations** - the mission phases and what each demands.
- **4. Trade space and figures of merit** - the design variables and how options are compared.
- **5. Technical budgets** - delta-v and mass (and others), with contingency and margins.
- **6. Risk register** - the scored risks and the mitigations for the serious ones.
- **7. Verification plan** - the table of how each requirement will be shown to hold.

Quick formulas

Risk score	$R = L \times C \quad (\text{likelihood} \times \text{consequence})$
Margin	$\text{Margin} = \frac{\text{capability} - \text{requirement}}{\text{requirement}} \times 100\%$
Budget with contingency	$\text{Total} = \left(\sum_i x_i\right)(1 + \text{contingency})$

Key glossary

- **Mission statement:** one or two sentences capturing what the mission does and for whom.
- **Requirement:** a verifiable shall-statement with a number and a condition.
- **ConOps:** the concept of operations, the mission story across its phases.
- **Technical budget:** a running total of a scarce resource (delta-v, mass) with contingency.
- **Margin:** the headroom between what the design delivers and what the requirement demands.

Going further

For the systems-engineering side, see the NASA Systems Engineering Handbook (SP-2016-6105) and Wertz and Larson, Space Mission Analysis and Design, for mission definition and budgeting. These are real, widely used references; please confirm editions yourself, as I cannot verify those details from here.